

Knowledge Capsule – Reinforcement Learning

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Day 3, TRAIL Data, AI and Multimodal Traffic Management

2025, Nov. 17th

Reinforcement Learning Framework

*How to perform **sequential decision-making** within the **environment** to maximize the **objective in the long run**?*

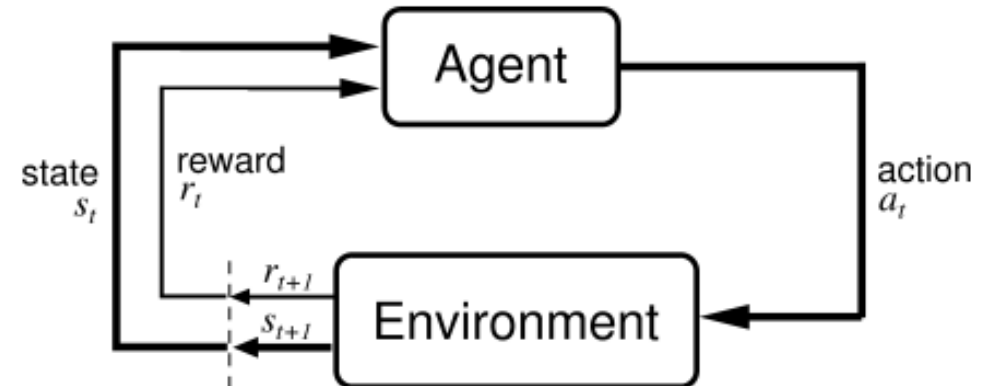
Agent and environment interact at discrete time steps : $t = 0, 1, 2, \dots$

Agent observes state at step t : $s_t \in S$

produces action at step t : $a_t \in A(s_t)$

gets resulting reward : $r_{t+1} \in \mathcal{R}$

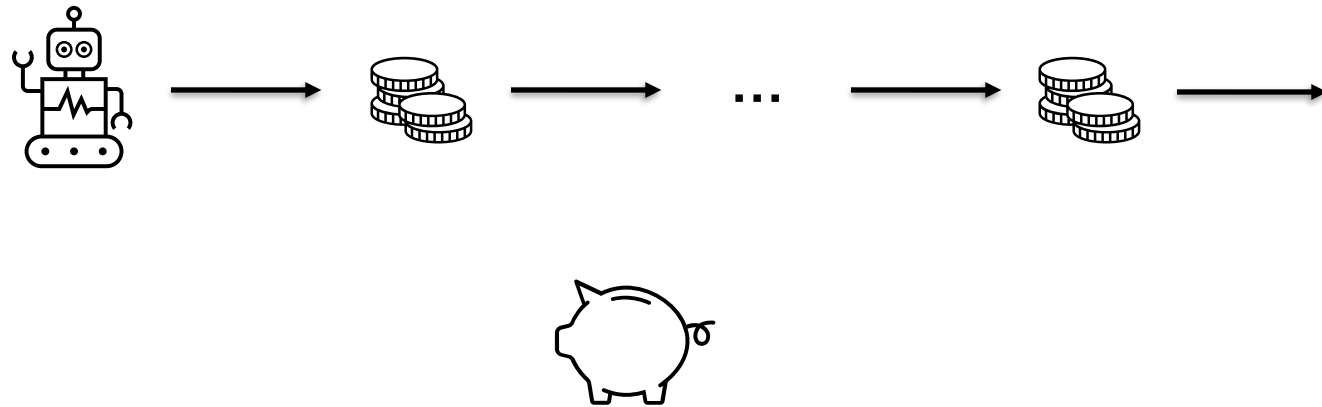
and resulting next state : s_{t+1}



We observe **trajectory** denoted by, $S_0, A_0, R_1, S_1, A_1, R_2, S_2, A_2, R_3, \dots$

Reward and Return

- **Reward** is immediately received at step t , denoted by r_t .
- **Return** R_t is defined as the cumulative reward from step t onwards.
- Our goal is to maximize the **expected return** $E(R_t)$



Long-term performance versus Immediate Reward Design

- How to translate the long-term objectives into actionable, immediate rewards for each decision?

*Strategic tradeoff between:
returns on the long-term objectives and short-term/ immediate rewards*

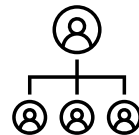
**Operator Overview:
Long-Term System Performance**



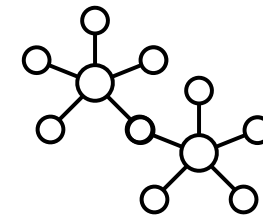
**Agent Execution:
Local, Step-wise Decision**



Global



Locals



Deep Reinforcement Learning

Incorporating deep learning with RL, **(multi-agent) deep reinforcement learning (DRL)** significantly improves the ability of agents to learn and make decisions in complex environments. The introduction of DRL leads to breakthrough performance in many areas, including **cooperative control**.

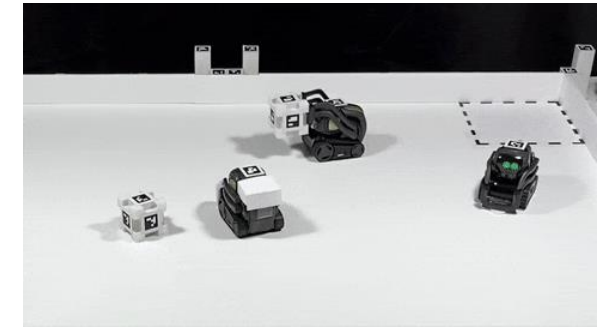
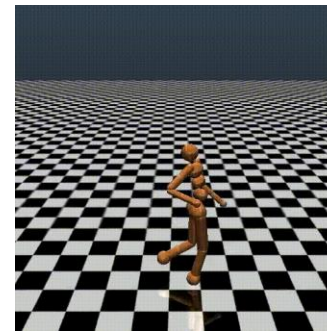
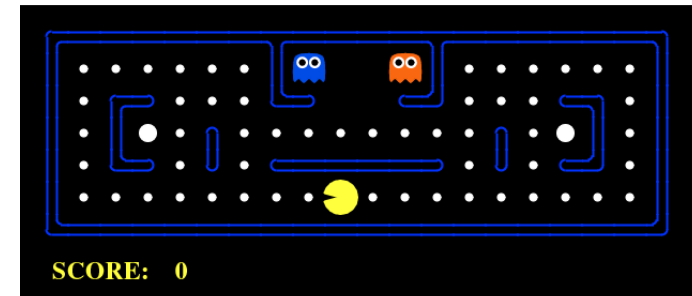
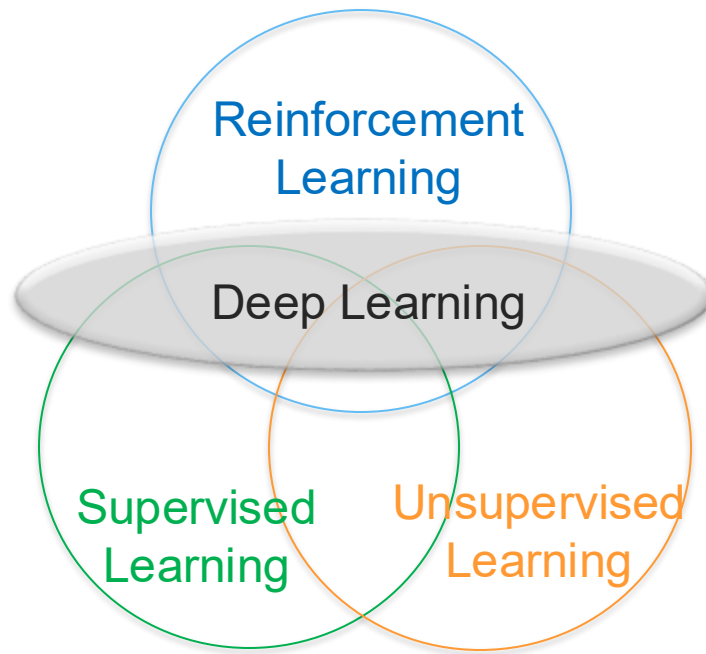


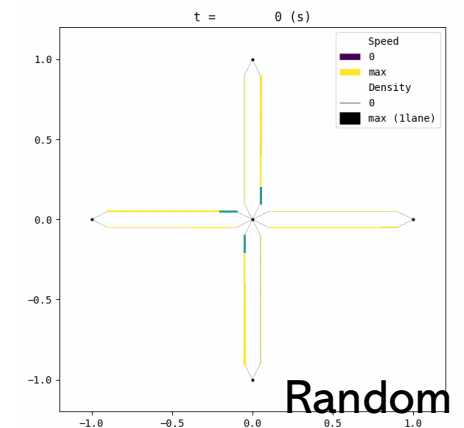
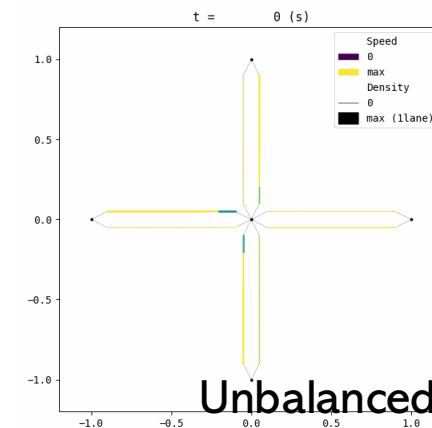
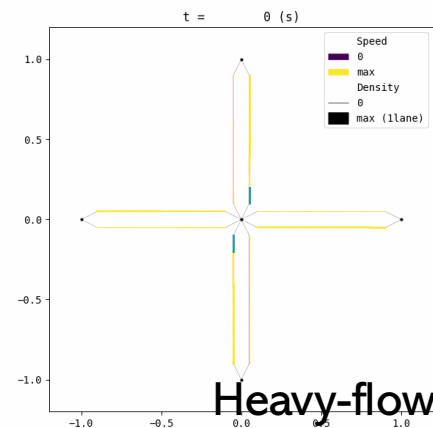
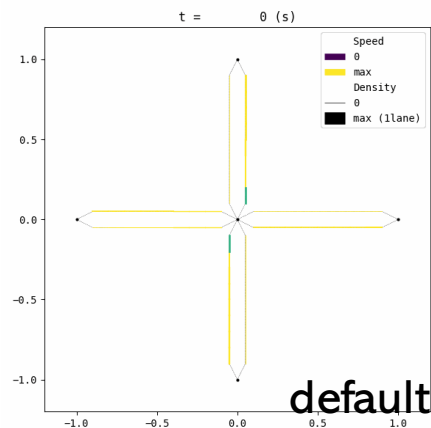
Figure Credit:

- Humanoid - Amid Fish, Lessons Learned Reproducing a Deep Reinforcement Learning Paper, 2018
- Pacman - Berkeley University, Intro to AI, CS188
- Cooperating robots with throw-and-lift tricks - Wu, Jimmy, et al. "Spatial intention maps for multi-agent mobile manipulation." 2021 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2021.

Reward design assignment for traffic light control

- ❑ In the exercise, you will apply the **reinforcement learning** and **traffic management** principles discussed in the lectures to a real-world scenario: controlling traffic lights at an intersection.
- ❑ The primary focus is on **designing an effective reward function** that guides the RL agent toward learning an optimal policy.
- ❑ A well-crafted reward function should encourage behaviors that improve long-term intersection performance: such as, minimizing average vehicle delay/ total travel time.

Demand Scenarios



Reward design assignment for traffic light control

Candidate Components for Reward Function

- Speed
- Density
- Flow
- Number of vehicles
- Queue length (number of vehicles on queue)
- Free-flow speed
- Jam density
- Capacity-in
- Capacity-out



[Link to Google Colab
Exercise Notebook](#)



Reading material for those who are interested

- Sutton, R. S., & Barto, A. G. (2018). *Reinforcement learning: An introduction*. MIT press.
- [Neural networks and Deep Learning explained by 3blue1brown](#)
- [Gymnasium \(Gym\) library by OpenAI](#)
- [AI Learns to Park - Deep Reinforcement Learning](#)
- [Two AI Fight for the same Parking Spot](#)
- [Champion-level Drone Racing using Deep Reinforcement Learning](#)

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